SOP-MECH-014 — Bearing Replacement — Horizontal Centrifugal Pumps (Sulzer CP Series)

Rev 3 | Effective 2023-02-01 | 12 steps | 240 minutes total

Procedure	SOP-MECH-014	**Revision**	Rev 3
Effective date	2023-02-01	**Review due**	2026-02-01
Applies to	P-101, P-102, P-105 (Sulzer CP series)	**Estimated duration**	240 minutes
Owner	D. Krishnan, Maintenance Manager	**Hold points**	6

1. Purpose

To define the controlled method for replacing the drive-end and non-drive-end rolling element bearings on Sulzer CP-series horizontal centrifugal pumps, such that the machine is returned to service within OEM vibration acceptance criteria and the bearing life stated in the OEM manual is achieved.

2. Scope and competency

Applies to P-101, P-102 and P-105. Execution requires a competent person certified on Sulzer CP-series bearing replacement. The certification is held on the training register; at the date of this revision one technician holds it.

3. References

- Sulzer CP 150-400 IOM, Section 7.4 — bearing replacement criteria (85% wear limit).
- RCA-2022-0825 — root cause analysis of the 2022-08-20 seizure.
- Plant permit-to-work procedure PTW-GEN-002.
- Lock-out / tag-out procedure SAF-ELE-006.

4. Personal protective equipment

- Safety helmet, safety spectacles, steel toe-capped footwear, coverall.
- Cut-resistant gloves for bearing handling; heat-resistant gloves for induction fitting.
- Hearing protection while the adjacent machine is running.

5. Tools and equipment

#	Item
1	Hydraulic bearing puller, 20 t
2	Induction bearing heater, 110 degC controlled
3	Dial indicator alignment set (rim and face)
4	Micrometer set, 75-100 mm, calibrated
5	Torque wrench, 40-200 Nm, calibrated
6	Vibration data collector with 1x/2x spectral capability

6. Safety notes

1. Hot work permit is NOT valid for bearing fitting. Flame heating destroys the bearing metallurgy and is prohibited.
2. The casing holds boiler feed water at up to 165 degC. Confirm drain-down and cool-down before breaking any joint.
3. Lock-out/tag-out at the 6.6 kV switchboard must be verified by a second person and recorded.
4. Lube oil is a slip hazard; bund the work area before disconnecting oil lines.

7. Procedure

Step	Instruction	Minutes	Hold point
1	Raise and validate the work permit. Confirm electrical isolation of the 250 kW driver at the 6.6 kV switchboard and apply lock-out/tag-out.	20	YES
2	Isolate suction valve HV-1012 and discharge non-return valve CV-1013. Drain the casing to the oily water sewer and confirm zero pressure at PI-1015.	20	
3	Disconnect the lube oil supply and return lines from LP-101A at the bearing housings. Blank the open ends.	10	
4	Record coupling alignment readings (rim and face) before disturbing the coupling. These are the reference for step 11.	15	YES
5	Remove the coupling spacer and the NDE bearing housing end cover. Photograph the bearing in situ before extraction.	20	
6	Extract the NDE angular-contact bearing (SKF 7314 BECBM) with a hydraulic puller. Do not apply load through the rolling elements.	30	
7	Measure and record the shaft journal diameter and the housing bore. Reject the shaft if the journal is more than 0.03 mm below nominal.	15	YES
8	Inspect the removed bearing and record the wear pattern against the OEM replacement criteria in Section 7.4 (85% limit). Retain the bearing for failure analysis if wear exceeds the limit.	15	
9	Induction-heat the replacement bearing to 110 degC maximum and fit it to the shaft. Never flame-heat and never drive a bearing on cold.	25	YES
10	Refit the bearing housing, renew the lip seals, and reconnect the lube oil lines. Confirm free rotation by hand.	20	
11	Re-align the pump to driver within 0.05 mm rim and 0.03 mm face, using the step 4 readings as the reference.	30	YES
12	Restore lubrication, remove isolation, and run for 30 minutes. Record vibration at the NDE and DE bearings; acceptance is below 4.5 mm/s RMS.	20	YES

Total estimated duration 240 minutes (4 hours) excluding permit issue and cool-down. 6 hold points require the supervisor's signature before the next step may start.

8. Hold points

Step	Hold point	Signatory
1	Raise and validate the work permit.	Maintenance supervisor
4	Record coupling alignment readings (rim and face) before disturbing the coupling.	Maintenance supervisor
7	Measure and record the shaft journal diameter and the housing bore.	Maintenance supervisor
9	Induction-heat the replacement bearing to 110 degC maximum and fit it to the shaft.	Maintenance supervisor
11	Re-align the pump to driver within 0.	Maintenance supervisor
12	Restore lubrication, remove isolation, and run for 30 minutes.	Maintenance supervisor

9. Acceptance criteria

- Overall vibration below 4.5 mm/s RMS at both bearing housings after 30 minutes running.
- Bearing housing temperature stable below 75 degC at rated duty.
- Lube oil header pressure at or above 1.4 bar throughout.
- Alignment within 0.05 mm rim and 0.03 mm face, recorded on the job card.
- No oil leakage from the housings after two hours of operation.

10. Records

- Completed job card with alignment readings before and after.

- Removed bearing wear assessment against the OEM Section 7.4 criteria.
- Post-job vibration record with 1x/2x spectrum.
- Photographs of the bearing in situ before extraction.

Revision history: Rev 1 2018-06-01 initial issue. Rev 2 2020-11-15 induction heating mandated. Rev 3 2023-02-01 — hold points added and wear assessment against the OEM limit made mandatory following RCA-2022-0825.